

MATHEMATICS
Introduction to Modern Algebra I
Solutions to Quiz 10.

1) Write $(1, 2, 5, 6)(6, 7, 2, 3)(4, 8)$ as a product of disjoint cycles. What is its sign (even or odd) and what is its order?

$(1, 2, 5, 6)(6, 7, 2, 3)(4, 8) = (1, 2, 3)(5, 6, 7)(4, 8)$. This is an odd permutation; its order is 6.

2) Write the inverse of $(1, 4, 3, 6)$ and write $(1, 4, 3, 6)$ as a product of transpositions.

$(1, 4, 3, 6)^{-1} = (1, 6, 3, 4)$ and $(1, 4, 3, 6) = (1, 6)(1, 3)(1, 4)$.

3) Let A_n be the subgroup of elements of even sign in S_n . Show that A_n is normal in S_n . (You do not have to show that A_n is a subgroup; just that this subgroup is normal!)

If $\sigma \in A_n$ and $g \in S_n$, then $\text{sgn}(\sigma) = 1$ and so $\text{sgn}(g^{-1}\sigma g) = \text{sgn}(g^{-1})\text{sgn}(\sigma)\text{sgn}(g) = \text{sgn}(g)^{-1} \cdot 1 \cdot \text{sgn}(g) = 1$, i.e. $g^{-1}\sigma g \in A_n$.